a gene, it is called as **homozygous(TT)**. An individual with two different alleles is called **heterozygous(Tt)**. Mendels non-true breeding plants are heterozygous, called as **hybrids**.

When the gene has two alleles the dominant allele is symbolized with capital letter and the recessive with small letter. When both alleles are recessive the individual is called **homozygous recessive** (tt) dwarf pea plants. An individual with two dominant alleles is called **homozygous dominant** (TT) tall pea plants. One dominant allele and one recessive allele (Tt) denotes non-true breeding tall pea plants **heterozygous tall.**

2.2.4 Mendelian inheritance – Mendel's Laws of Heredity

Mendel proposed two rules based on his observations on monohybrid cross, today these rules are called laws of inheritance The first law is The Law of Dominance and the second law is The Law of Segregation. These scientific laws play an important role in the history of evolution.

The Law of Dominance: The characters are controlled by discrete units called factors which occur in pairs. In a dissimilar pair of factors one member of the pair is dominant and the other is recessive. This law gives an explanation to the monohybrid cross (a) the expression of only one of the parental characters in F_1 generation and (b) the expression of both in the F_2 generation. It also explains the proportion of 3:1 obtained at the F_2

The Law of Segregation (Law of Purity of gametes): Alleles do not show any blending, both characters are seen as such in the F_2 generation although one of the characters is not seen in the F_1 generation. During the formation of gametes, the factors or alleles of a pair separate and segregate from each other such that each gamete receives only one of the two factors. A homozygous parent produces similar gametes and a heterozygous parent produces two kinds of gametes each having one allele with equal proportion. **Gametes are never hybrid.**

2.3 Monohybrid cross

Monohybrid inheritance is the inheritance of a single character i.e. plant height.It involves the inheritance of two alleles of a single gene. When the F_1 generation was selfed Mendel noticed that 787 of 1064 F_2 plants were tall, while 277 of 1064 were dwarf. The dwarf trait disappeared in the F_1 generation only to reappear in the F_2 generation. The term **genotype** is the genetic constitution of an individual. The term **phenotype** refers to the observable characteristic of an organism. In a genetic cross the genotypes and phenotypes of offspring, resulting from combining gametes during fertilization can be easily understood with the help of a diagram called Punnett's Square named after a British Geneticist Reginald C.Punnett. It is a graphical representation to calculate the probability of all possible genotypes of offsprings in a genetic cross.The Law of Dominance and the Law of Segregation give suitable explanation to Mendel's monohybrid cross.

Reciprocal cross – In one experiment, the tall pea plants were pollinated with the pollens from a true-breeding dwarf plants, the result was all tall plants. When the parental types were reversed, the pollen from a tall plant was used to pollinate a dwarf pea plant which gave only tall plants. The result was the same - All tall plants. Tall (♀) x Dwarf (♂) and Tall (♂) x Dwarf (♀) matings are done in both ways which are called reciprocal crosses.The results of the reciprocal crosses are the same. So it was concluded that the trait is not sex dependent. The results of Mendel's monohybrid crosses were not sex dependent.

The gene for plant height has two alleles: Tall (T) x Dwarf (t). The phenotypic and genotypic analysis of the crosses has been shown by Checker board method or by Forkline method.

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